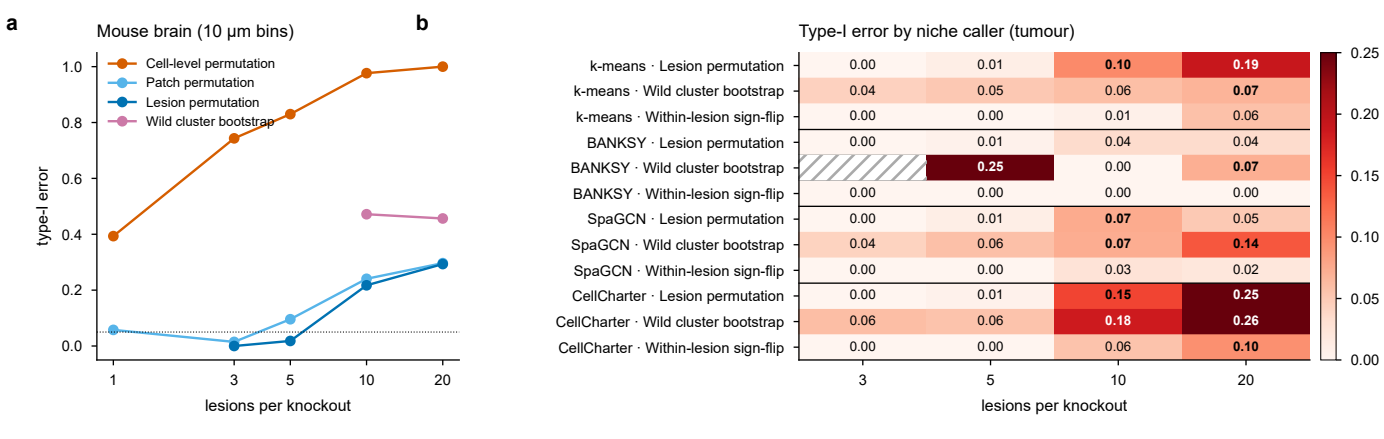

a
Mouse brain (10 µm bins)
Cell-level permutation
Patch permutation
Lesion permutation
Wild cluster bootstrap
type-I error
1.0
0.8
0.6
0.4
0.2
0.0
1
3
5
10
20
lesions per knockout
b
Type-I error by niche caller (tumour)
k-means · Lesion permutation 0.00 0.01 0.10 0.19
k-means · Wild cluster bootstrap 0.04 0.05 0.06 0.07
k-means · Within-lesion sign-flip 0.00 0.00 0.01 0.06
BANKSY · Lesion permutation 0.00 0.01 0.04 0.04
BANKSY · Wild cluster bootstrap 0.25 0.00 0.07
BANKSY · Within-lesion sign-flip 0.00 0.00 0.00 0.00
SpaGCN · Lesion permutation 0.00 0.01 0.07 0.05
SpaGCN · Wild cluster bootstrap 0.04 0.06 0.07 0.14
SpaGCN · Within-lesion sign-flip 0.00 0.00 0.03 0.02
CellCharter · Lesion permutation 0.00 0.01 0.15 0.25
CellCharter · Wild cluster bootstrap 0.06 0.06 0.18 0.26
CellCharter · Within-lesion sign-flip 0.00 0.00 0.06 0.10
3
5
10
20
lesions per knockout
0.25
0.20
0.15
0.10
0.05
0.00